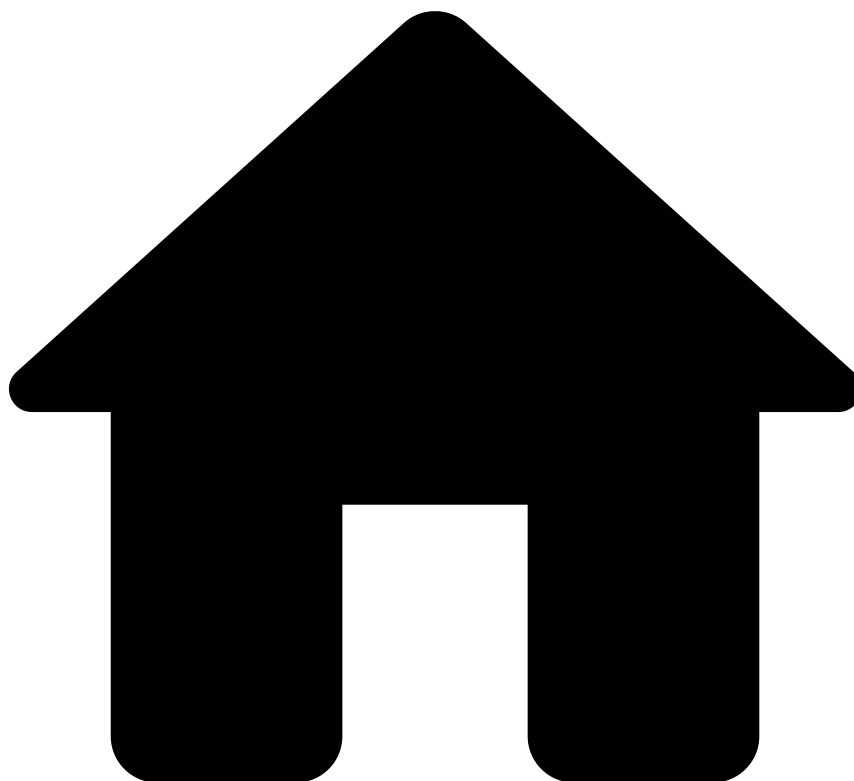


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On this page

Analyze the AutoML Output

Was this page helpful? 

The AutoML output analysis allows gives you an insight on the end results, i.e. if they have fulfilled your business needs or if you should continue with data science iterations. This page walks you through an example of an AutoML run output analysis.

You can find the output of an AutoML run in the **Data Science** tab on the following panels:

- [Data sources](#): Includes the 'Feature Generation Results' data source.
- [Plans](#): Includes the plan that has the definition of the new features.

- [Model Projects](#): Includes the recommended model.

How to analyze an AutoML run

An AutoML run generates various resources including data sources, plans and model projects. However, if you want to quickly assess the quality of the results, focus on the model project output. More specifically, use the following available options:

- **ROC curve**: Check how to analyze the performance of the recommended model in the [Model evaluations](#) section.
- **Feature Importance**: Identify the subset of features that contribute the most to the recommended model in the [Model evaluations](#) section.
- **Hyperparameter results**: See the combination of different parameters used to train various models in the [Hyperparameter tunings](#) section.

AutoML output

This section provides a description for the output of each resource generated by AutoML. Note that, as mentioned in the [How to analyze an AutoML run](#) section, the most relevant results that define the quality of the AutoML run are described in the [Model projects](#) section.

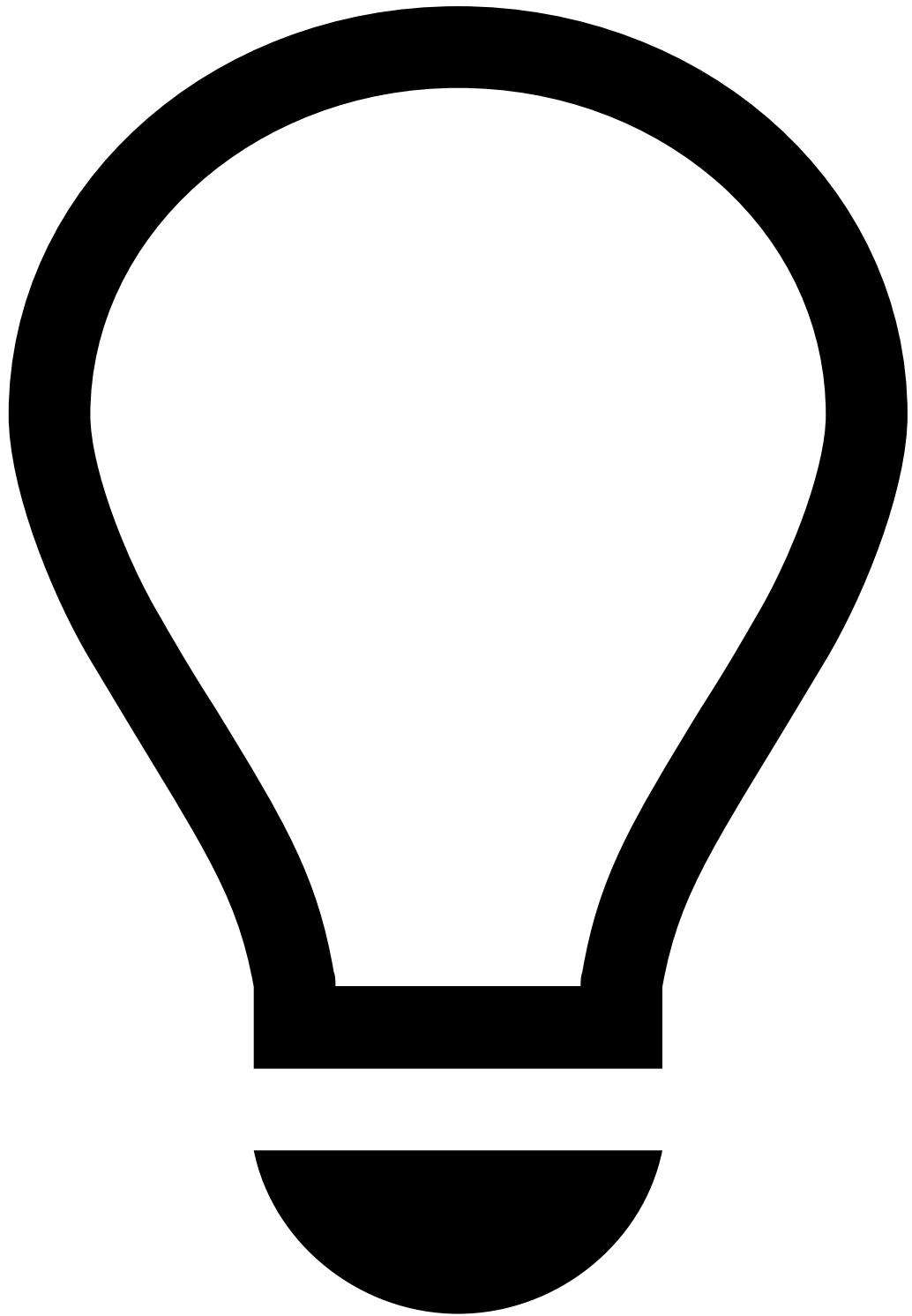
Data sources

When executing an AutoML run, Pulse processes the selected input data source and creates new features. Along with this process, Pulse generates and makes the following data sources available:

- **Feature Generation Results**: Contains the data enriched with the new features.
- **Train dataset / Validation dataset / Test dataset**: Individual data sources that contain data corresponding to training, validation and test time ranges, as defined during the AutoML setup.

You can use the generated data sources to quickly explore the results:

1. Open the **Data Science** tab and select a data source (e.g. 'Feature Generation Results').
2. Select the button to open the **Data Source Analysis** menu:
3. Select the **Field available for breakdown**, i.e. a field corresponding to the fraud label.
4. Select the **Analyse** button to compute the statistics.



tip

Breakdowns: The breakdown option allows you to select a field to later use as a filter for your analysis. Consider that you select the 'fraud label' field and compute statistics for the 'fraud' and 'not fraud' breakdown values. Once the analysis job is computed, you can open the results of each feature and view the data associated to the 'fraud' or the 'not fraud' labels individually.

5. Once the statistics are computed, you can use the **View Statistics** button to view the results for each feature. You can see the data distribution for numerical and categorical values. You can use this functionality to perform quick validations - for example, to confirm that the metrics related to sum of amounts are not negative.

Plans

During an AutoML run, Pulse generates the following plans:

- **Datasource Split: Non-deployable plan** that defines which part of the input data source is used for training, validation and, optionally, testing the data science models. You can open the plan to see the time ranges that were selected in the setup of the AutoML run.
- **Feature Generation: Deployable plan** that contains all the fields that are generated during the AutoML run. The operators inside this plan are described in the [Feature Generation plan](#) section.
- **Feature Generation Reduced: Deployable plan** that contains only the relevant features selected by the AutoML during the feature selection step.



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Ready for deployment: Both feature generation and reduced plans can be directly integrated in the runtime workflow:

- Use the feature generation plan if you want to get the most of all the features generated by AutoML.
- Use the feature generation reduced plan if you want to integrate it with the model recommended by AutoML.

Feature Generation plan

The following image shows an example of a feature generation plan created by an AutoML run:



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Operators and their features definition: The number of operators - and hence number and quality of the features - depends significantly on how the fields are tagged in the semantic file. Tagging the input data fields incorrectly may not produce meaningful features (or features at all). Note that the quality of models depends on the features, and consequently on model results, including ROC curve and metrics, depend heavily on the tagging quality.

The map and metric operators in this plan are explained as follows:

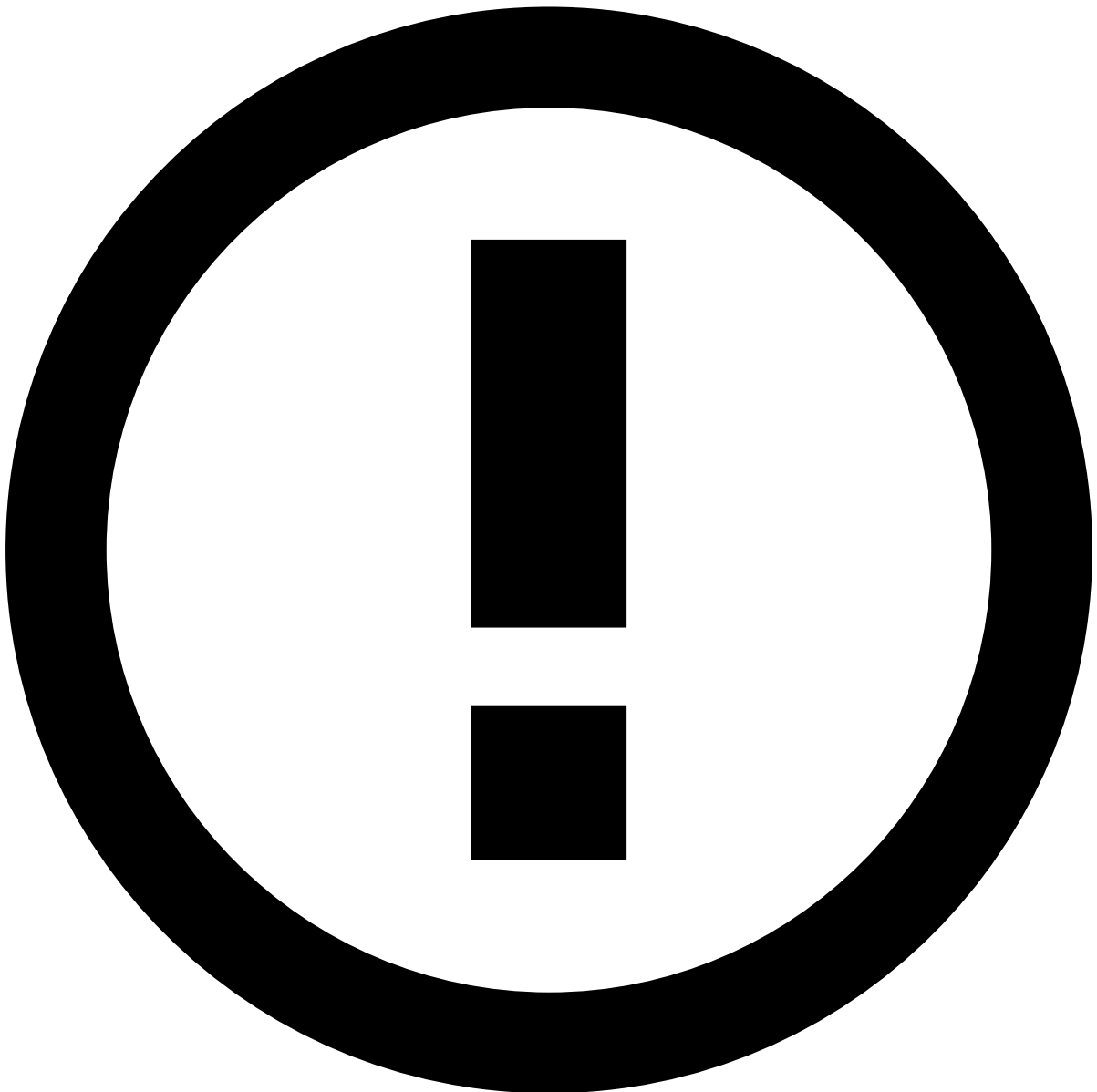
- **Feature 1:** Contains basic data transformations and data normalizations such as:
 - Extracting the month day from a given field, e.g. the month of the account open date.
 - Calculating the time difference between the `event_occurred_at` and the `account_open_date` fields.

- **Profiles:** Contains the field definition that represents the behavior of a given entity, such as card or merchant, in a specific type frame (i.e. also known as segments-of-one profiles). Examples include:
 - Average amount spent by a given card in the last month.
 - Number of transactions in the last month.
- **Feature 2:** Contains calculations performed over the fields defined in the profiles operator, such as:
 - Ratio between two fields, e.g. a certain count per hour divided by the count per day.

Remember that this output is used as an example. The number and type of operators depend on semantic tags and on the input data source, and therefore, your plan may have different operators.

Model projects

For each run, AutoML stores all the information related to models in a new model project.



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Ready for deployment: AutoML recommends the best model, which can be directly integrated in the runtime workflow.

The following image shows an example of a model project for an AutoML run:

You can quickly assess the performance of the trained models with:

- [Model Evaluations](#): Check the ROC curve and Feature Importance results.
- [Hyperparameter Tunings](#): Check the results of testing different sets of parameters.

Model evaluations

Inside the model project, open the model evaluation generated by AutoML to check the ROC curves, metrics and Feature Importance:

1. Navigate to the **Data Science** tab and open the model project generated by AutoML.
2. Under the **Model Evaluations** table, select the one that you want to analyze.
3. Check the following subsections to learn how to analyze ROC curves, metrics and Feature Importance.

ROC curve

The ROC curve shows the recall (percentage of events that are fraud in the real world, and that were classified as fraud by the risk engine) for a given false-positive rate (FRP - percentage of events that are not fraud in the real world, and that were classified as fraud by the risk engine). An example of a good metric indicator is a recall above 70% at an FPR of 1%. You can use the range options to easily check this value by narrowing down the axis range:

Metrics

You can also see a list of metrics associated with the recommended model. You can use the **Threshold** bar to visualize the metrics at a certain value, e.g. at a FPR of 1%:

Feature Importance

To access the Feature Importance results, you may have to compute it first. Inside the model evaluation, use the button to run the job. Once the job is finished, a new tab is displayed in the **Model Evaluation** panel with a list of features that impact the model the most:

Hyperparameter tunings

An AutoML run trains several models using a hyperparameter optimization method:

1. Navigate to the **Data Science** tab and open the model project generated by AutoML.
2. Select the tuning result under the **Hyperparameter Tunings** table.
3. Select the **Exploration Mode** tab. The following figure shows an example of the hyperparameter tuning result at a 1.0% FPR:

Next step

Read the [Integrate Resources in Runtime](#) to start scoring transactions in real time.